

LOGIC No: SEQ-3401	DESCRIPTION: CATALYST REDUCTION REACTOR (DC-3401A) + HEATER EA-3401 LOGIC	SIL: SIL 2	TAG: DC-3401A FLOC: TJC-LLD-3400-01
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CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP HEATER EA-3401	EFF-2 CLOSE H2 INJECTION FV-3403	EFF-3 OPEN REACTOR VENT PV-3404	EFF-4 MAINTAIN / INCREASE N2 FV-17343	EFF-5 ANNUNCIATE ALARM ON DCS
T1	Bed temperature HIGH-HIGH	TSHH-3401	> 230 degC	2oo3	X	X		X	X
T2	Reactor pressure HIGH-HIGH	PSHH-3404	> 5 barg	1oo2		X	X		X
T3	Outlet O2 HIGH during H2 (inerting fail)	AI-3401	> 100 ppm	1oo1		X		X	X
T4	Nitrogen carrier flow LOW-LOW	FSL-17343	< 150 kg/h	1oo1	X	X			X
T5	Heater element over-temperature	EHT-3401	skin TC > 260 degC	1oo1	X				X
T6	Emergency stop (manual)	HS-3401	Manual	1oo1	X	X	X		X

Legend: X = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP HEATER EA-3401
EFF-2	CLOSE H2 INJECTION FV-3403
EFF-3	OPEN REACTOR VENT PV-3404
EFF-4	MAINTAIN / INCREASE N2 FV-17343
EFF-5	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	O2 free / inerting done (AI-3401 < 100 ppm)	AI-3401
2	N2 carrier flow established (> 150 kg/h)	FT-17343
3	Bed homogeneous 120 degC (all TE within 15 degC)	TE-3401
4	No active trip / reset done	DCS reset
=>	ALL permissives TRUE (AND) => START DC-3401A ENABLED	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID TJC-LLD-PID-3401 for instrument loops and the Datasheet for design limits of DC-3401A.